
INVENTORY MANAGEMENT MODEL FOR RAW MATERIALS AT *NORITAKE (PVT) LTD*

OPTIMIZING THE ORDER QUANTITY &
TIME FOR THE REPLENISHMENT



Acknowledgment

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We would like to express our special gratitude to our colleagues for their kind co-operation and encouragement which help us to complete this project.

Abstract

This is our MT 325 course project report. This is done to increase the efficiency of ordering the raw materials of the NORITAKE (PVT) LTD in Matale. In such company planning order quantity and time that between two orders are must. And when communicating with management we understood that they need a planning for their ordering cycle that minimize the total inventory cost.

Inventory management theory provides a more efficient way of planning the orders of raw materials with production according to the given data. Most of the time this used when order or produce and keep the products lot in a store and sales with the demand, but not like the most cases, here we had to do order keep lot of raw materials and order those.

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1. Introduction

1.1 Company Overview



Noritake Lanka Porcelain Private Limited is the pioneer porcelain tableware manufacturing conglomerate in Sri Lanka. The Company was established in 1972 as a joint venture company between Ceylon Ceramic Corporation and Noritake Co., Limited of Japan. Noritake Co. Limited of Japan is a hundred years old company which markets tableware under the brand "Noritake" all over the world.

The Company's production facility is located in Matale where country's rich minerals such as quartz, feldspar and dolomite deposits are spread. Factory operates with the production systems developed by the parent company in Japan. There are several Japanese technical advisors station in the factory to monitor production process to ensure products are manufactured in par with the international quality standards. The factory started its operations in 1972 just with 150 workforce and as at today workforce stands with 1200 skilled workers. Most of the workers, supervisors and managers of the factory are trained in Japan in their relevant fields.

Since the inception, company went with several expansion programs by investing nearly Rs.1000 million to increase the production facility from 3000 sets to 12000 sets per month. In year 2008 the Company invested Rs.160 million to introduce Bone China Tableware making plant. Bone China Tableware is regarded as the most expensive tableware in the world and we will be the pioneer in this segment in Sri Lanka.

Tableware products manufactured by the Company are well within the standards issued by FDA of the United State of America for lead and cadmium release. This standard requires that the maximum level of lead and cadmium release to be at 0.5% PPM. We maintain this level below 0.2% PPM. Lead and Cadmium are small particles coming out from the decoration materials used to decorate tableware at the usage. These partials can cause cancer in human body if it reaches more than 0.5% PPM.

Special features of their products

- Very high whiteness – 90%
- High translucency – 0.12%
- High Thermal shock resistance – 180 c
- Body is completely vitreous
- High temperature fired – 1270c
- High scratch resistance
- Resistant to acid and alkalis

Maintenance of all these features made possible by the facts that product designing, selection and processing of raw materials, processing of printing materials such as pigment and gold, silk printing screens are all done within Noritake enterprise.

Apart from Japanese expertise, our management processes are supported by ISO 9001:2000 certification from the Sri Lanka Standard Institute. Their factory in Matale is one of the largest production facility in the world to make high quality tableware under the brand "**Noritake**".

1.2 Production Process

Raw materials like *Quartz*, *Feldspar* and *Kaolin* are crushed in a ball mill for 36 hours and the resultant liquid is dewatered and kept as a cake for maturing under ideal humidity and temperature. The resultant liquid is dewatered and kept as a cake for maturing under ideal humidity and temperature.

The seasoned cake is then driven through a mill to create clay cylinders used to sculpt the product. The Noritake factory is equipped with machines for both automatic and manual formation whether it is done manually or automatically.



Forming a porcelain object is an Art and Art cannot be without human intervention. Producing a cup is not complete until the edge trimming and handle attachments are done. The handle needs to be firmly pasted and accurately positioned. The same applies to the construction of openings and attaching spouts and handles on a teapot. Only small and versatile feminine hands and patience with a suitable temperament can perform these exacting tasks with high precision. Sponging and finishing too require a gentle feminine touch.



Every stage of production is closely supervised by Japanese experts. Before going to the next stage of the process - the soft biscuit firing, the finished greenware is subjected to close inspection. The journey through this kiln moulds the greenware into ideal water absorbing conditions for glazing. The greenware is heated at [950](#) degrees Celsius, until it is fit to go to the glaze bar. The fired soft biscuit is inspected again for any damage before being sent for brushing and glazing. Skilled hands are required again at this stage of the process. Firmly holding a plate by the fingertips while dipping it into the recommended depth in liquid glaze is not a task for all. The machine is only to transport the glazed item while the glazing is done manually.

After sponging the edge or the foot, the glazed item is driven through a kiln for another round of heating called glossed firing. It is during this journey through the flames at a temperature of [1300](#) degrees Celsius that the item receives its shine. The heated item is exposed to another test which is known as the pipeware inspection. This inspection rules out any possibility of passing a blemished item beyond this point.



Then comes the decal application. Noritake Lanka porcelain uses both decalcomania and traditional hand painting techniques. The decalcomania technique is well suited for mass production. Patterns that are printed on paper are transferred to the piece, pasting the dampened paper with a pattern on the surface. Patience and concentration are essential human qualities needed at this stage to maintain the accuracy of the placement of the pattern.

Then it is exposed to a third round of firing known as decal firing. The paper is burnt and the pattern with over glazed colors is transferred to the surface of the piece. Hand painting by skilful artists gives a touch of uniqueness to each product. The gold lining is another task that requires skill. Delicate fingertips are well trained to move the pointer brush along a perfect

path. This brownish coat converts to gold once burnt. The end product is finally inspected at this stage before being sent for packaging. No room is left in the production process for slipping in an inferior item to the market.

Noritake Lanka is confident and competent enough to guarantee the same quality that Noritake Japan has been maintaining through hundreds of years. It is a blend of superior technology and dedicated craftsmanship enriched by an enduring history.



Noritake Factory

1.3 Problems We Found in Production Process

When we observe the production process, we found that they have perfect planning schedule for the overall production process except for the raw material inventory.

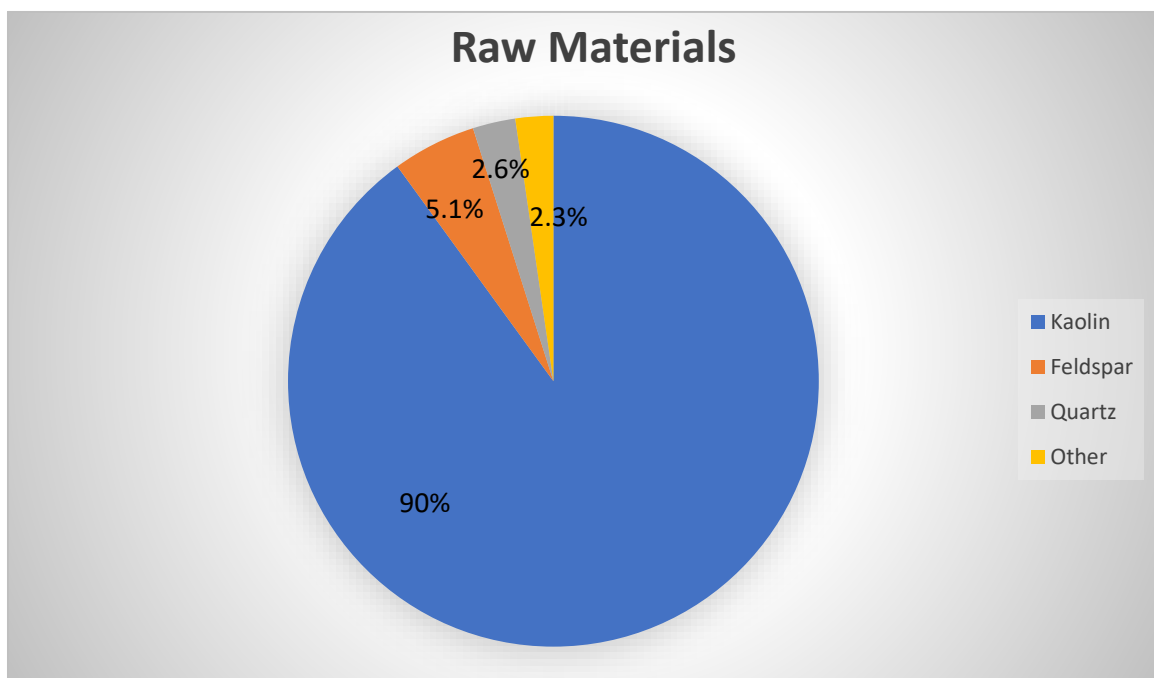
As Mr. Duminda Herath (Production Engineer) mentioned, they order the raw materials when the lot is decreasing and quantity of each raw material that order is not predefined. Since they wish to minimize the total inventory cost we have consider about the following facts,

1. How much to order from each raw material
2. When to order from each raw material

Those minimize the total inventory cost of raw materials.

So we decide to develop a model to find the most economic order quantity and optimal time for the replenishment that minimize the total raw material inventory cost.

Since they use three major raw materials such as Kaolin, Feldspar and Quartz in following percentages, we need to develop models for each raw material.



Graph 1.1

2. Methodology

2.1 Inventory Management

The word “Inventory” means a physical stock of goods which is kept in hand for smooth and efficient running of current or future affairs of an organization.

E.g.- The stock of raw materials before product

The stock of finished good after production

Inventory can be classified as follows.

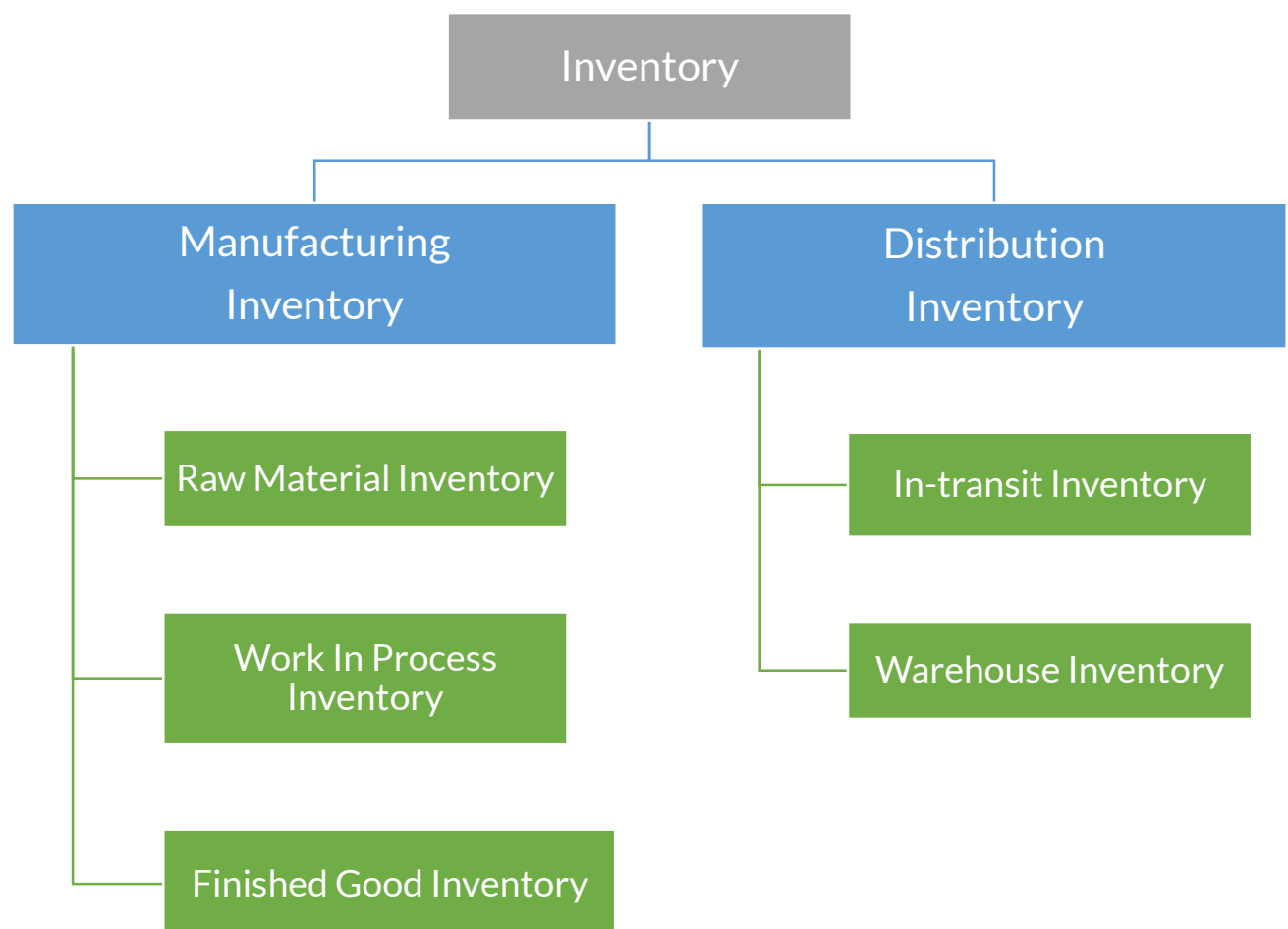


Figure 2.1

The various cost terms as economic parameter associated with inventory problems are as follows.

1. Carrying Cost / Holding Cost

This cost includes the cost of maintaining the inventory. The cost is directly proportional to the quantity to be kept in stock and the time to which the inventory is maintain. It includes the following costs.

- Taxes
- Insurances
- Spoilage
- Theft
- Obsolescence
- Salaries & wages warehouse employees
- Utilities & building cost for warehouse
- Supplies such as forms & papers for the warehouse

It is generally denoted by C_h per unit of quantity, per unit of time

2. Ordering / Replenishment Cost

This is the cost incurred the replenish the inventory or say changing the inventory. It is generally denoted by C_o and it is not proportional to the quantity to be ordered.

In order to define the total inventory cost we have to consider following factors,

1. Demand (D)

Demand is an economic principle that describes a consumer's desire and willingness to pay a price for a specific good or service. Holding all others factors constant, an increase in the price of a good or service will decrease demand, and vice versa.

2. Order Quantity (q)

The amount or number of a material in a order.

3. Cycle Time (t)

The time period that each quantity (q) is produced or supplied.

4. Unit Time (T)

Entire time period that total demand (D) is required. Basically,

$$T = 1 \text{ yr}$$

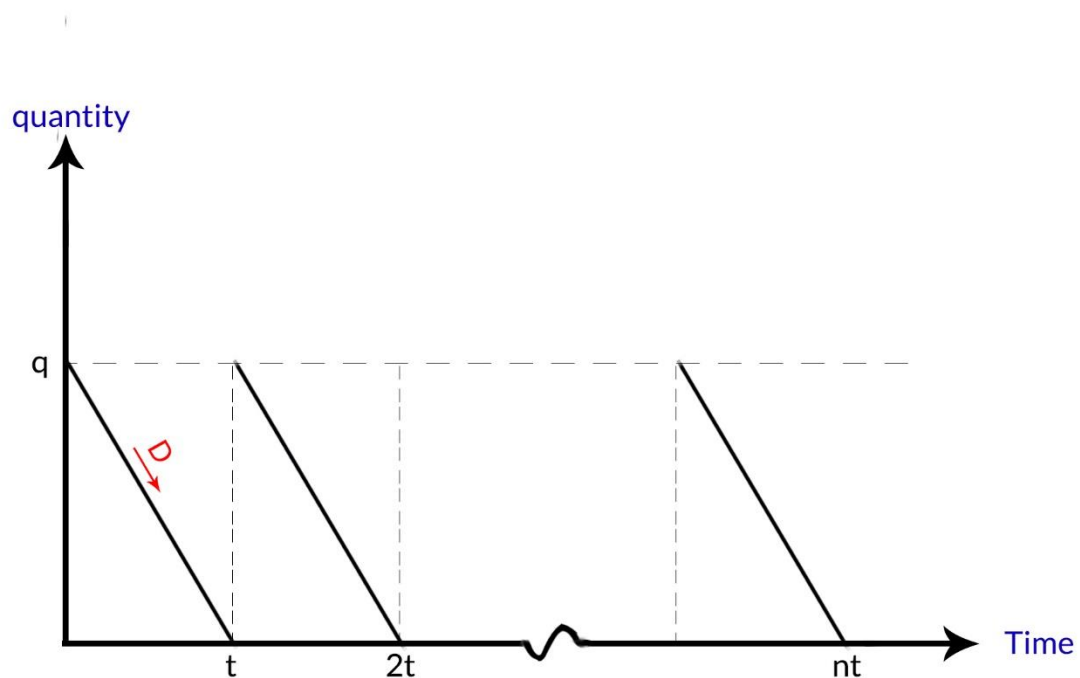
5. No of Cycles(n)

Total number of orders that require in the unit time (T).

5.2 Consumption Model

Assumptions

1. Demand is known and it at uniform rate D units of quantity per unit of time (T). In such cases we say that the demand is continuous.
2. Shortages are not permitted. That is as soon as the level of the inventory reaches zero the inventory is replenished. Thus the cost shortages is assuming to be zero.
3. The rate of replenishment of the inventory is infinity. That is the production or supply to the inventory is instantaneous.
4. Lead time is zero.
5. The order quantity (lot size) is same for each cycle.



Graph 2.1

Objective

Here we have to find the optimal value of q which is economic. That is minimized the total inventory cost. For that we use following formula derived using graph 2.1.

The total Inventory in one cycle (t time & one order) is given by,

$$\frac{1}{2} * t * q = \text{area of the triangle}$$

The total order cost for one cycle is given by,

$$\frac{C_o}{t}$$

Therefore the total cost for the one cycle is given by,

$$C_h \left(\frac{1}{2} tq \right) + \frac{C_o}{t}$$

Thus the total annual cost is given by,

$$C(q) = \frac{\frac{1}{2} tq C_h}{t} + \frac{C_o}{t}$$

$$nt = 1$$

$$n = \frac{1}{t} = \frac{D}{q}$$

$$C(q) = \frac{1}{2} q C_h + \frac{D}{q} C_o$$

$$\frac{d C(q)}{dq} = 0 \Rightarrow$$

$$\frac{C_h}{2} - \frac{D C_o}{q^2} = 0$$

$$q_o = \sqrt{\frac{2 D C_o}{C_h}}$$

$$\frac{d^2C(q)}{dq^2} > 0 \quad (\text{at } q = q_o)$$

Thus the total cost minimize at q_o

Economic Order Quantity (EOQ)

$$q^* = \sqrt{\frac{2DC_o}{C_h}}$$

$$\frac{D}{q} = \frac{1}{t}$$

$$t = \frac{q}{D}$$

$$t^* = \frac{q^*}{D}$$

$$t^* = \sqrt{\frac{2DC_o}{C_h}} * \frac{1}{D}$$

$$t^* = \sqrt{\frac{2C_o}{DC_h}}$$

6. Data & Results

As we requested production engineer gave us a brief description about how production goes. There are two type of items produced in the factory. Those are

1. Flatware



This mean flat items like plates, trays. Those are produced using pressing machine & mould. Simply we can describe the process such as, put the clay mixture on the mould and press by the machine.

2. Hollowware



This mean items with a hollow like cups, jugs. These type items produced by using mould with hollow and two walls. Fill the hollow from sloppy clay mixture. Then after the specific time strain the mixture and then item create remaining clay layer in the mould.

When we collect the data we got the data separately for each item types.

For our objective we need the following data.

1. Annual Demand of the raw materials (D)
2. Ordering cost per order (C_o)
3. Holding cost per unit per year (C_h)

3.1 Annual Demand (D)

We have to find the EOQ (q^*)s and t^* s of the three main raw materials. Therefore we need the annual demand of the Kaolin, Feldspar and Quartz. For that we request the production data of last few years. They gave us to production data of the 2014, 2015 and 2016. Those data are in separate table for two item types Flatware and Hollowware.

We collect the monthly used amount of three materials in each item type. This amount contain materials used in whole production process. This mean to produce greenware and moulds and in the glazing process.

Usage of the Raw Materials (In metric tons)												
Kaolin				Felsphar				Quartz				
Month	Flatware	Hollowware	Total	Month	Flatware	Hollowware	Total	Month	Flatware	Hollowware	Total	
2014/01	12.29	14.47	26.76	2014/01	0.62	0.70	1.32	2014/01	0.22	0.33	0.55	
2014/02	12.33	13.76	26.09	2014/02	0.43	0.58	1.01	2014/02	0.29	0.43	0.72	
2014/03	12.36	10.78	23.14	2014/03	0.62	0.84	1.46	2014/03	0.24	0.39	0.63	
2014/04	8.61	11.91	20.52	2014/04	0.67	0.82	1.49	2014/04	0.29	0.37	0.66	
2014/05	10.83	13.74	24.58	2014/05	0.33	0.62	0.95	2014/05	0.11	0.43	0.53	
2014/06	8.99	12.68	21.66	2014/06	0.67	0.74	1.41	2014/06	0.29	0.34	0.64	
2014/07	10.79	11.81	22.61	2014/07	0.39	0.86	1.25	2014/07	0.12	0.32	0.44	
2014/08	10.89	12.93	23.82	2014/08	0.59	0.69	1.28	2014/08	0.27	0.40	0.67	
2014/09	9.47	10.41	19.88	2014/09	0.67	0.78	1.45	2014/09	0.33	0.32	0.65	
2014/10	12.40	12.13	24.53	2014/10	0.40	0.86	1.26	2014/10	0.14	0.39	0.53	
2014/11	10.00	13.76	23.77	2014/11	0.29	0.74	1.03	2014/11	0.17	0.45	0.62	
2014/12	10.89	10.85	21.74	2014/12	0.39	0.62	1.01	2014/12	0.19	0.44	0.63	
2015/01	9.81	10.67	20.48	2015/01	0.49	0.88	1.37	2015/01	0.13	0.39	0.52	
2015/02	9.29	11.97	21.25	2015/02	0.67	0.80	1.47	2015/02	0.13	0.33	0.46	
2015/03	10.76	14.48	25.25	2015/03	0.48	0.85	1.33	2015/03	0.21	0.42	0.63	
2015/04	11.87	11.73	23.61	2015/04	0.28	0.58	0.87	2015/04	0.35	0.42	0.77	
2015/05	9.56	14.18	23.75	2015/05	0.30	0.81	1.11	2015/05	0.28	0.29	0.57	
2015/06	10.62	10.94	21.56	2015/06	0.32	0.83	1.15	2015/06	0.11	0.40	0.51	
2015/07	10.64	10.61	21.25	2015/07	0.39	0.89	1.28	2015/07	0.30	0.32	0.62	
2015/08	12.43	10.86	23.30	2015/08	0.45	0.73	1.19	2015/08	0.10	0.29	0.39	
2015/09	9.53	12.80	22.34	2015/09	0.62	0.90	1.52	2015/09	0.15	0.28	0.43	
2015/10	12.52	11.95	24.47	2015/10	0.66	0.72	1.38	2015/10	0.20	0.37	0.57	
2015/11	8.77	10.78	19.55	2015/11	0.40	0.74	1.14	2015/11	0.21	0.41	0.62	
2015/12	9.29	11.21	20.50	2015/12	0.25	0.87	1.12	2015/12	0.30	0.41	0.71	
2016/01	12.44	12.87	25.32	2016/01	0.21	0.63	0.84	2016/01	0.34	0.37	0.71	
2016/02	8.83	11.66	20.48	2016/02	0.54	0.71	1.25	2016/02	0.23	0.28	0.51	
2016/03	12.21	13.38	25.59	2016/03	0.61	0.63	1.24	2016/03	0.28	0.42	0.70	
2016/04	11.43	11.57	23.00	2016/04	0.48	0.76	1.25	2016/04	0.17	0.38	0.56	
2016/05	9.87	11.01	20.88	2016/05	0.65	0.63	1.29	2016/05	0.14	0.45	0.59	
2016/06	8.83	12.47	21.30	2016/06	0.29	0.81	1.10	2016/06	0.26	0.43	0.69	
2016/07	12.35	12.96	25.32	2016/07	0.54	0.88	1.42	2016/07	0.11	0.37	0.49	
2016/08	8.64	10.41	19.05	2016/08	0.54	0.83	1.37	2016/08	0.17	0.40	0.57	
2016/09	10.26	11.77	22.03	2016/09	0.30	0.57	0.88	2016/09	0.18	0.40	0.59	
2016/10	12.13	13.14	25.27	2016/10	0.53	0.63	1.16	2016/10	0.22	0.31	0.52	
2016/11	10.96	13.94	24.90	2016/11	0.59	0.61	1.20	2016/11	0.32	0.32	0.64	
2016/12	10.01	11.77	21.78	2016/12	0.41	0.68	1.09	2016/12	0.34	0.32	0.65	
			821.32				43.93				21.32	

Table 3.1

3.2 Ordering Cost (C_o)

We observed that ordering cost contains following terms. They gave monthly data for those costs.

1. Commissions
2. Broker charges
3. Others

We calculate Ordering cost per order for year 2016 for each raw material.

Currently they order kaolin at each week.

Ordering Cost for Kaolin					
Month	Commission(Rs)	Broker Charges(Rs)	Other(Rs)	Total(Rs)	
2016/01	16000	12000	3000	31000	
2016/02	16000	12000	4500	32500	
2016/03	16000	12000	1800	29800	
2016/04	16000	12000	2550	30550	
2016/05	16000	12000	4450	32450	
2016/06	16000	12000	8210	36210	
2016/07	16000	12000	1970	29970	
2016/08	16000	12000	5460	33460	
2016/09	16000	12000	3550	31550	
2016/10	16000	12000	2150	30150	
2016/11	16000	12000	780	28780	
2016/12	16000	12000	4200	32200	
Total	192000	144000	42620	378620	

Table 3.2

Currently they order Feldspar & Quartz one time per month.

Ordering Cost for Feldspar				
Month	Commission(Rs)	Broker Charges(Rs)	Other(Rs)	Total(Rs)
2016/01	4000	2000	2175	8175
2016/02	4000	2000	1960	7960
2016/03	4000	2000	2325	8325
2016/04	4000	2000	2050	8050
2016/05	4000	2000	2100	8100
2016/06	4000	2000	2600	8600
2016/07	4000	2000	2110	8110
2016/08	4000	2000	1840	7840
2016/09	4000	2000	1510	7510
2016/10	4000	2000	2245	8245
2016/11	4000	2000	2300	8300
2016/12	4000	2000	2070	8070
Total	48000	24000	25285	97285

Table 3.3

Ordering Cost for Quartz				
Month	Commission(Rs)	Broker Charges(Rs)	Other(Rs)	Total(Rs)
2016/01	4000	2000	1950	7950
2016/02	4000	2000	1680	7680
2016/03	4000	2000	1990	7990
2016/04	4000	2000	1710	7710
2016/05	4000	2000	1930	7930
2016/06	4000	2000	1850	7850
2016/07	4000	2000	1945	7945
2016/08	4000	2000	2020	8020
2016/09	4000	2000	1450	7450
2016/10	4000	2000	1850	7850
2016/11	4000	2000	2130	8130
2016/12	4000	2000	1755	7755
Total	48000	24000	22260	94260

Table 3.4

3.3 Holding Cost per Unit per time

They have data for maintain the warehouse. All three materials are kept in the same warehouse. We collect the following data. These are for three years.

Cost for the ware house				
Year	Insurance	Salary & Wages	Utilities	Total Cost
2014	400000	1260000	180000	1840000
2015	400000	1488000	195000	2083000
2016	400000	1716000	210000	2326000
	1200000	4464000	585000	6249000

Table 3.5

Salary & wages contain salary for the labors, warehouse managers and security. They gave us the monthly data. We calculate for the year. Utilities contain electricity bill, repairing cost, stationary cost. This total cost come for the all three materials.

3.4 Current Inventory Cost

Annual total inventory cost is given by,

$$C_h \left(\frac{1}{2} tq \right) + \frac{C_o}{t}$$

Here when we calculate the inventory cost we have used mean value for the order quantity (q) and for the C_o (calculated at next part) and C_h , have used observed values by previous data.

1. Kaolin

$$\begin{aligned} q &= 821.32 / 108 \\ &= 7.6 \text{ tons} \end{aligned}$$

$$\begin{aligned} \text{Annual inventory cost} &= 7048.59 * 0.02 * 7.6 * 0.5 + \frac{7887.92}{0.02} \\ &= 394931Rs \end{aligned}$$

2. Feldspar

$$\begin{aligned} q &= 43.93 / 36 \\ &= 1.22 \text{ tons} \end{aligned}$$

$$\begin{aligned} \text{Annual inventory cost} &= 7048.59 * 0.08 * 1.22 * 0.5 + \frac{8107.08}{0.08} \\ &= 101682.47Rs \end{aligned}$$

3. Quartz

$$\begin{aligned} q &= 21.32 / 36 \\ &= 0.59 \text{ tons} \end{aligned}$$

$$\begin{aligned} \text{Annual inventory cost} &= 7048.59 * 0.08 * 0.59 * 0.5 + \frac{7855}{0.08} \\ &= 98353.85Rs \end{aligned}$$

3.5 Economic Order Quantity (EOQ)

$$q^* = \sqrt{\frac{2DC_o}{C_h}}$$

To find the EOQ for each raw material, we calculate Demand (D), Ordering cost (C_o) and Carrying cost / Holding cost (C_h)

According to the table 3.5 we can observe the Holding cost (C_h) for the all materials,

$$\begin{aligned} \text{Annual Total Holding cost} &= \frac{6249000}{3} \\ &= 2083000 \end{aligned}$$

$$\begin{aligned} \text{Total numbers of units per year} &= \frac{821.32+43.93+21.32}{3} \\ &= 295.52 \end{aligned}$$

$$\begin{aligned} \text{Holding cost per unit per unit time} &= \frac{2083000}{295.52} \\ C_h &= 7048.59 \end{aligned}$$

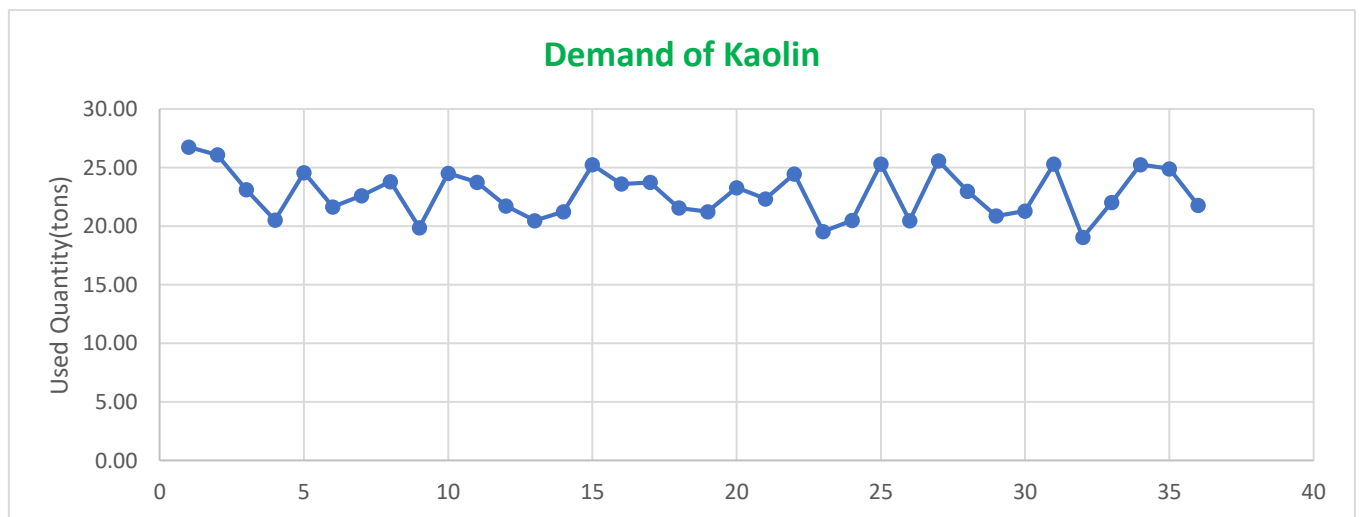
Other than these three few other materials in the warehouse. But their size is too small respect to the main materials. Thus we ignore that when calculate the holding cost.

3.5.1 Economic Order Quantity for Kaolin

$$D = \frac{821.32 \text{ t}}{3}$$

$$D = 273.77 \text{ t}$$

To use above formula which is explained in the chapter 2.2, D should be uniform for the whole time interval. We considered about data of last three years.



Graph 3.1

According to the above graph clearly we can see demand of the kaolin is uniform. Therefore we can use above describe formula.

According to the table 3.2 we can observe the Ordering cost (C_o) for the kaolin,

$$\begin{aligned} \text{Number of Order in the 2016} &= 4 * 12 \\ &= 48 \end{aligned}$$

Thus,

$$C_o = \frac{378620}{48}$$
$$= 7887.92 \text{ Rs}$$

By previous part,

$$C_h = 7048.59$$

Now we can observe the EOQ and t^*

$$q^* = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2 * 273.77 * 7887.92}{7048.59}}$$

$$q^* = 24.75 \text{ ton}$$

$$t^* = \sqrt{\frac{2C_o}{DC_h}} = \sqrt{\frac{2 * 7887.92}{273.77 * 7048.59}}$$

$$= 0.09 \text{ yrs}$$

$$t^* = 32.85 \text{ days}$$

Now Total annual inventory cost for the Kaolin,

$$C_{ka} = C_h \left(\frac{1}{2} tq \right) + \frac{C_o}{t} = 7048.59 * 0.09 * 24.75 * 0.5 + \frac{7887.92}{0.09}$$

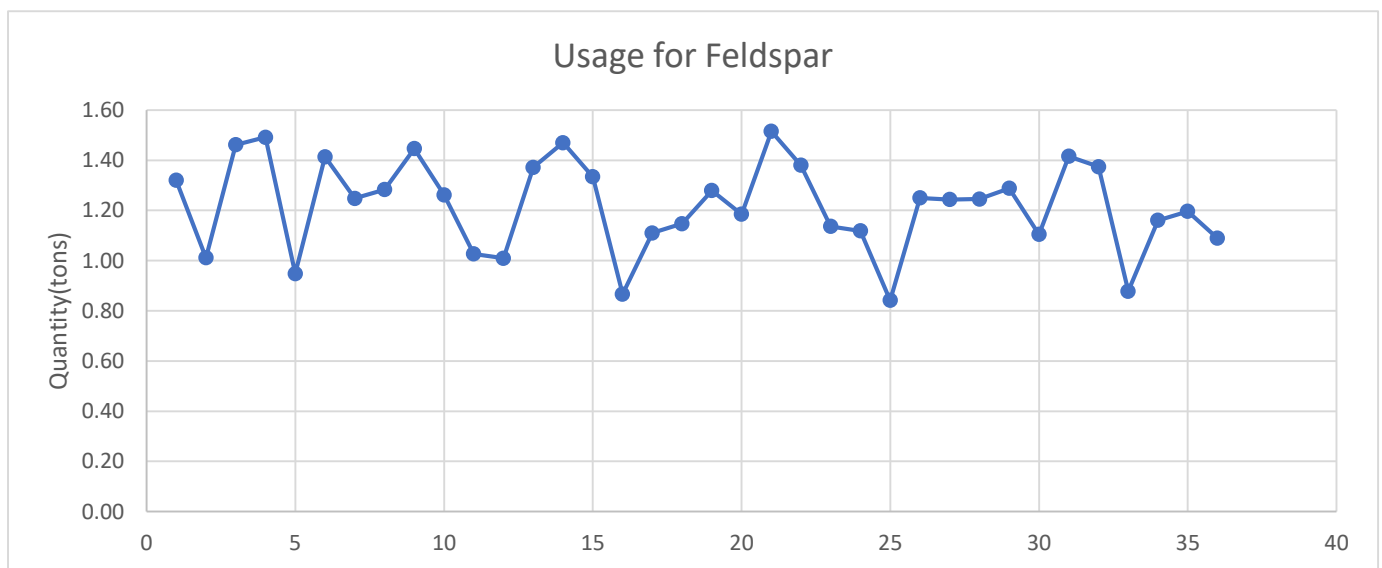
$$C_{ka} = 95493.92 \text{ Rs}$$

3.5.2 Economic Order Quantity for Feldspar

$$D = \frac{14.64}{3}$$

$$D = 4.88 \text{ tons}$$

To use above formula which is explained in the chapter 2.2, D should be uniform for the whole time interval. We considered about data of last three years.



Graph 3.2

According to the above graph clearly we can see demand of the kaolin is uniform. Therefore we can use above describe formula.

According to the table 3.3 we can observe the Ordering cost (C_o) for the Feldspar,

$$\text{Number of Order in the 2016} = 12$$

Thus,

$$C_o = \frac{97285}{12}$$

$$= 8107.08 \text{ Rs}$$

By previous part,

$$C_h = 7048.59$$

Now we can calculate the EOQ and t^*

$$q^* = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2 * 4.88 * 8107.08}{7048.59}}$$

$$q^* = 3.35 \text{ ton}$$

$$t^* = \sqrt{\frac{2C_o}{DC_h}} = \sqrt{\frac{2 * 8107.08}{4.88 * 7048.59}}$$

$$= 0.69 \text{ yrs}$$

$$t^* = 250.6 \text{ days}$$

Now Total annual inventory cost for the Feldspar,

$$C_{fe} = C_h \left(\frac{1}{2} tq \right) + \frac{C_o}{t} = 7048.59 * 0.69 * 4.88 * 0.5 + \frac{8107.08}{0.69}$$

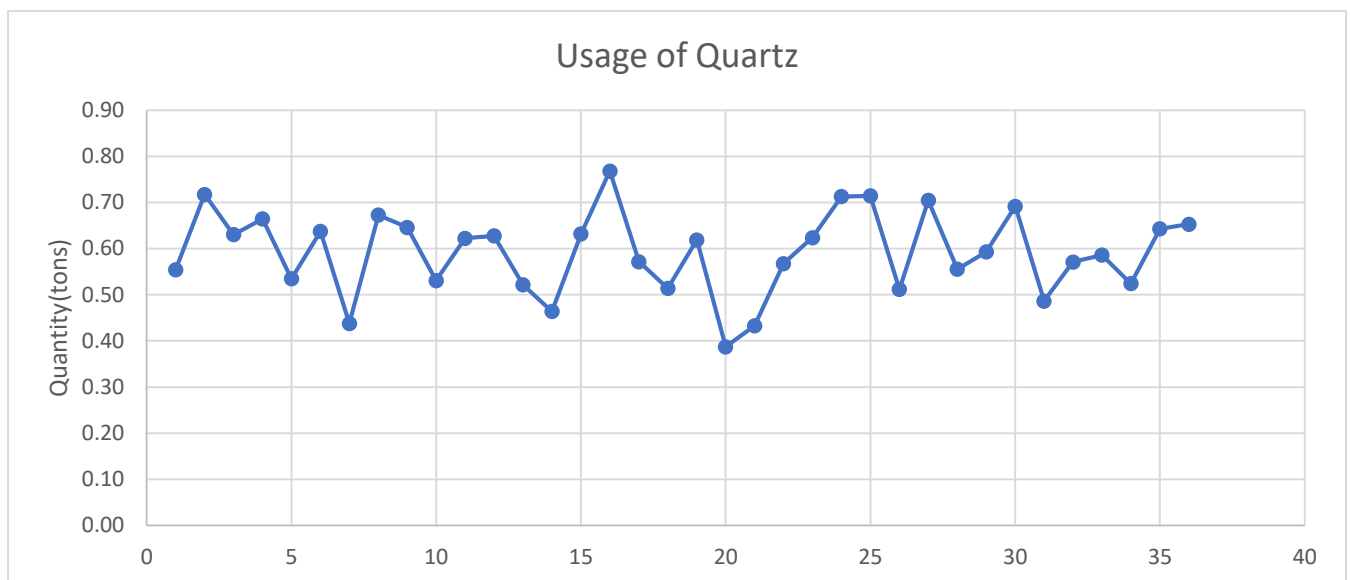
$$C_{fe} = 23616.39Rs$$

3.5.3 Economic Order Quantity for Quartz

$$D = \frac{7.1 \text{ t}}{3}$$

$$D = 2.37 \text{ t}$$

To use above formula which is explained in the chapter 2.2, D should be uniform for the whole time interval. We considered about data of last three years



Graph 3.3

According to the above graph clearly we can see demand of the kaolin is uniform. Therefore we can use above describe formula.

According to the table 3.4 we can observe the Ordering cost (C_o) for the Feldspar,

$$\text{Number of Order in the 2016} = 12$$

Thus,

$$C_o = \frac{94260}{12}$$
$$= 7855 \text{ Rs}$$

By previous part,

$$C_h = 7048.59$$

Now we can calculate the EOQ and t^*

$$q^* = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2 * 2.37 * 7855}{7048.59}}$$

$$q^* = 2.3 \text{ ton}$$

$$t^* = \sqrt{\frac{2C_o}{DC_h}} = \sqrt{\frac{2 * 7855}{2.37 * 7048.59}}$$

$$= 0.97 \text{ yrs}$$

$$t^* = 353.96 \text{ days}$$

Now Total annual inventory cost for the Quartz,

$$C_{qu} = C_h \left(\frac{1}{2} tq \right) + \frac{C_o}{t} = 7048.59 * 0.97 * 2.37 * 0.5 + \frac{7855}{0.97}$$

$$C_{qu} = 16199.94 \text{ Rs}$$

4. Conclusions & Discussion

4.1 Conclusions

When we look at their ordering plans, we understood they don't try to minimize the inventory cost. They just bring materials weekly or monthly.

Following table shows current inventory costs and costs which occur if they follow our formula.

Materials	Current Cost	Cost from Our model
Kaolin	394931 <i>Rs</i>	95493.92 <i>Rs</i>
Feldspar	101682.47 <i>Rs</i>	23616.39 <i>Rs</i>
Quartz	98353.85 <i>Rs</i>	16199.94 <i>Rs</i>

Table 4.1

According to the above table clearly we can see noticeable decrease in the inventory cost of the all three materials.

Some time there are small differences in these values because at some points we use mean values. But there is a noticeable advantage if use our model even above error is at the calculation.

4.2 Suggestions

We explain to NORITAKE (PVT) LTD management, plan the ordering raw materials perfectly. Then they can increase the profit by minimize the inventory cost of raw materials. We suggest to them use our model at least for one materials for one year. Then they can check the accuracy of it and identify what are the points of the model should be improved. Then they can find the optimum plan for the ordering plan which minimize the inventory cost of raw materials.

Also we suggest to them, consider the inventory cost of produced items. If that is very complicate, we prefer the take from help another research team. Then they can maximize the total profit by minimize the inventory cost of products.

4.3 Difficulties

Since we are the external team of the company, we faced some troubles when collecting the data.

Analysis the data that we collect was very difficult to us. We had to go different sections and meet varies peoples for collect the data.

If we could stay at the factory for few weeks, maybe we can develop this model perfectly. With other works we faced some difficult. For any work, we had to go to Matale from here. It was very difficult with other works.

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